IT3011 - Theory & Practices in Statistical Modelling

Population Proportions

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$(1-\alpha)100\%$ Confidence Interval for population proportion of success for large samples

Consider a population in which the proportion of success is p. If a random sample of n is taken from this population and then number of successes in the sample (X) allows the binomial distribution.

$$X \sim Bin(n, p)$$

Now for large n,

$$X \sim N(np, npq) \quad where\ q = 1 - p$$

Now consider $\hat{P} = \frac{X}{n}$. Then,

$$E(\hat{P}) = p \quad and \quad V(\hat{P}) = \frac{pq}{n}$$

So, for large samples ($n > 30$),

$$\hat{P} \sim N(p, \frac{pq}{n})$$

$(1-\alpha)100\%$ Confidence Interval for population proportion of success for large samples

By standardizing this statistic we get that,

$$Z = \frac{\hat{P} - p}{\sqrt{\frac{pq}{n}}} \sim N(0,1)$$

Then we can construct a $(1-\alpha)100\%$ Confidence Interval for p as follows.

$$P\left(-Z_{\frac{\alpha}{2}} \leq Z \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(-Z_{\frac{\alpha}{2}} \leq \frac{\hat{P} - p}{\sqrt{\frac{pq}{n}}} \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

So the confidence interval will be,

$$\left[\hat{P} - Z_{\frac{\alpha}{2}}\sqrt{\frac{pq}{n}} \quad , \qquad \hat{P} + Z_{\frac{\alpha}{2}}\sqrt{\frac{pq}{n}}\right] \approx \left[\hat{P} - Z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}\hat{q}}{n}} \quad , \qquad \hat{P} + Z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}\hat{q}}{n}}\right]$$

$(1-\alpha)100\%$ Confidence Interval for population proportion of success for large samples

Ex-

A manufacturer wants to assess the proportion of defective items in a large batch produced by a particular machine. A random sample of 300 items is taken and it is observed that 45 items of them are defective. Calculate a 95% confidence interval for the proportion of defective items in the complete batch.

$(1-\alpha)100\%$ Confidence Interval for population proportion of success for large samples

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A manufacturer wants to assess the proportion of defective items in a large batch produced by a particular machine. A random sample of 300 items is taken and it is observed that 45 items of them are defective. Calculate a 95% confidence interval for the proportion of defective items in the complete batch.

$$\hat{p} = \frac{45}{300} = 0.15$$

$$\hat{q} = 0.85$$

$$\hat{P} \pm Z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}\hat{q}}{n}} = 0.15 \pm 1.96\sqrt{\frac{0.15 \times 0.85}{300}}$$

$$[0.11\ , 0.19]$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success for large samples

Consider we have taken two samples from two populations. Then for large sample sizes,

$$\hat{p}_1 \sim N\left(p_1, \frac{p_1 q_1}{n_1}\right)$$

$$\hat{p}_2 \sim N(p_2, \frac{p_2 q_2}{n_2})$$

Then,

$$(\hat{p}_1 - \hat{p}_2) \sim N\left(p_1 - p_2, \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}\right)$$

So,

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}} \sim N(0,1)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success for large samples

Then we can construct a $(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success as follows.

$$P\left(-Z_{\frac{\alpha}{2}} \leq Z \leq Z_{\frac{\alpha}{2}}\right) = 1-\alpha$$

$$P\left(-Z_{\frac{\alpha}{2}} \leq \frac{(\hat{p}_1-\hat{p}_2)-(p_1-p_2)}{\sqrt{\frac{p_1q_1}{n_1}+\frac{p_2q_2}{n_2}}} \leq Z_{\frac{\alpha}{2}}\right) = 1-\alpha$$

So the confidence interval will be,

$$\left[(\hat{p}_1-\hat{p}_2)-Z_{\frac{\alpha}{2}}\sqrt{\frac{p_1q_1}{n_1}+\frac{p_2q_2}{n_2}}, (\hat{p}_1-\hat{p}_2)+Z_{\frac{\alpha}{2}}\sqrt{\frac{p_1q_1}{n_1}+\frac{p_2q_2}{n_2}}\right] \approx$$

$$\left[(\hat{p}_1-\hat{p}_2)-Z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1}+\frac{\hat{p}_2\hat{q}_2}{n_2}}, (\hat{p}_1-\hat{p}_2)+Z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1}+\frac{\hat{p}_2\hat{q}_2}{n_2}}\right]$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success for large samples

Ex:-

A clinic trial is conducted to determine if a certain type of inoculation has an effect on the incident of a certain disease. A sample of 1000 rats was kept in a controlled environment for a period of 1 year and 500 rats were given the inoculation. Of the group not given the drug, there was 120 incidences of the disease, while 98 of the inoculated group contracted it. Compute the 90% confidence interval for the difference between the probability of incidents of the disease in uninoculated rats and the probability of incidents after receiving the drug.

$(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success for large samples

Ex:-

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$$\hat{p}_1 = \frac{120}{500} = 0.24 \quad \hat{p}_2 = \frac{98}{500} = 0.196$$

$$(\hat{p}_1 - \hat{p}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}} = 0.044 \pm 1.65 \times \sqrt{\frac{0.24 \times 0.76}{500} + \frac{0.196 \times 0.804}{500}} = 0.044 \pm 0.043$$

$$[0.001, 0.087]$$

Test involving one sample proportion

Consider following hypotheses,

$$H_0 : p = k \text{ , } H_1 : p \neq k$$

$$H_0 : p = k \text{ , } H_1 : p > k$$

$$H_0 : p = k \text{ , } H_1 : p < k$$

Where p represents the population proportion of success and $0 \leq k \leq 1$. Then the test statistic is,

$$Z = \frac{\hat{P} - p}{\sqrt{\frac{pq}{n}}} \sim N(0,1)$$

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From planted 100 seeds, 83 seeds were germinated . Test the manufacturers claim, the seeds are highly productive with 90% germination ratio. Use 5% level of significance.

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$$H_0 : p = 0.9 \text{ , } H_1 : p \neq 0.9$$

$$Z_{cal} = \frac{\hat{P} - p}{\sqrt{\frac{pq}{n}}} = \frac{0.83 - 0.9}{\sqrt{\frac{0.9 \times 0.1}{100}}} = -2.33$$

$$Z_{2.5\%} = 1.96 \ \ and \ \ -Z_{2.5\%} = -1.96$$

$$Z_{cal} < -1.96$$

So reject H_0 at 5% level of significance.

Conclusion: There is an evidence to say that the claim of the manufacturer is not true at 5% level of significance.

Test involving two population proportions

Consider following hypotheses,

$$H_0 : p_1 - p_2 = k \text{ , } H_1 : p_1 - p_2 \neq k$$

$$H_0 : p_1 - p_2 = k \text{ , } H_1 : p_1 - p_2 > k$$

$$H_0 : p_1 - p_2 = k \text{ , } H_1 : p_1 - p_2 < k$$

Where p represents the population proportion of success and $0 \leq k \leq 1$.

We know that,

$$(\hat{p}_1 - \hat{p}_2) \sim N\left(p_1 - p_2, \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}\right)$$

There are two different test statistics for this parameter, the choice of which are to be used depends on the value of **k**.

Test involving two population proportions

Case 01: $k = 0$

The test statistic is,

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1)$$

$$\hat{p}_1 = \frac{x_1}{n_1} \ (The\ proportion\ of\ success\ in\ sample\ 01)$$

$$\hat{p}_2 = \frac{x_2}{n_2} \ (The\ proportion\ of\ success\ in\ sample\ 02)$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2} \ (The\ common\ proportion\ of\ success\ in\ sample\ 01\ \&\ 02)\ and\ \hat{q} = 1 - \hat{p}$$

Test involving two population proportions

Case 02: $k \neq 0$

The test statistic is,

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}} \sim N(0,1)$$

Test involving two population proportions

Ex:-

In a manufacturing process, two different methods A and B were employed. To evaluate both of manufacturing processes, the manager decided to examine the differences between them. He took 3740 items from each process and 60 defectives were found in process A and 28 in process B. Test at 5% level whether process B is better than process A.

Test involving two population proportions

Ex:-

In a manufacturing process, two different methods A and B were employed. To evaluate both of manufacturing processes, the manager decided to examine the differences between them. He took 3740 items from each process and 60 defectives were found in process A and 28 in process B. Test at 5% level whether process B is better than process A.

$$H_0 : p_B - p_A = 0 \text{ , } H_1 : p_B - p_A < 0$$

$$\hat{p}_A = \frac{60}{3740} = 0.02 \quad \hat{p}_B = \frac{28}{3740} = 0.0075 \;\; \hat{p} = \frac{60+28}{7480} = 0.01 \;\; \hat{q} = 0.99$$

$$Z = \frac{(\hat{p}_B - \hat{p}_A) - (p_B - p_A)}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_B} + \frac{1}{n_A}\right)}} = \frac{(0.0075 - 0.02) - 0}{\sqrt{0.01 \times 0.99 \times \left(\frac{1}{3740} + \frac{1}{3740}\right)}} = -5.43$$

$$-5.43 < -Z_{5\%} = -1.65$$

So reject H_0 at 5% level of significance.

Conclusion : There is an evidence to say that the process B is better than the process A at 5% level of significance